

DIGITAL FORENSICS

MALICIOUS EMAIL ANALYSIS

5 May 2026

Connor Robison

Overview/Executive Summary.....	2
Objectives	3
Forensic Analysis	4
Detailed Steps Taken	4
Justification for Methods and Tools	4
Evidence:	5
Relevant Findings.....	10
Conclusion.....	11

Overview/Executive Summary

This report details the forensic examination of evidence from a disk image of an LGE Nexus 5 Android device obtained regarding the alleged illegal purchase of Owls for pet purposes. Before analyzing the image, we compared the hashes of the .raw image file against image_info.txt, an information file produced by Magnet imager at the time of the image acquisition. Both the MD5 hash and the SHA-1 hash produced by Magnet imager at the time of acquisition matched the hashes of the evidence image, proving the integrity of the source examined.

The strongest findings from the image examined are the following:

1. The device is associated with the user Sarah McAvoy, confirmed by two linked accounts including a google account (mcavoys87@gmail.com) and a skype account (mcavoys87). This proves that Sarah McAvoy is the user of this device.
2. Chrome browser search history contains research regarding owl legality and acquisition. Searches include but are not limited to: "Owls as Pets," "Where can I buy an owl?," Birdtrader owl listings, and exoticanimalsforsale.net searches for "Owl." Additionally, chrome history shows activity on amazon, where the user browsed listing for bird bedding products before the delivery related communications on January 30.
3. Skype messages between Sarah McAvoy and Matt Haze (generalhaze28) on January 30-31, 2017, containing suspicious language such as "thanks for the hook up" and references to meeting up.
4. On January 31, 2017, at 7:41 PM EST, the device received an SMS message from 304-518-4333 containing the following: "Sarah, the delivery is today 7 tonight the confirmation will come later through pidgin." Sarah responded 30 seconds later saying, "Thank you!"
5. On February 3, 2017, multiple owl related images were saved to the device, including two images of snowy owls.

Combined, these artifacts support the conclusion that the owner of the device imaged (Sarah McAvoy) knowingly researched owl acquisition, communicated with a seller, received messages related to delivery, and later saved images of owls.

Objectives

This forensic investigation was conducted to answer the following questions regarding the Nexus device:

1. Who was the user of the device?
2. Did the device contain evidence related to the research or illegal purchase of an owl?
3. Did the device contain artifacts related to other individuals in an illegal sale of an owl?
4. If there was evidence of a sale, what does the timeline look like?

Forensic Analysis

Detailed Steps Taken

1. The acquisition metadata from Magnet was reviewed to identify the source device and hashes at the time of imaging.
2. The MD5 and SHA-1 hashes of “LGE Nexus 5 Full Image.raw” were verified against the hashes in the Magnet metadata.
3. The image was loaded into Autopsy, and the default ingest modules were run, producing a clean output of communication accounts, contacts, installed programs, messages, web bookmarks, cookies, and chrome browser history.
4. Contacts, calls, messages, browser activity and downloads were inspected for potentially incriminating content
5. Relevant events were converted to EST, as the user is located in eastern time.

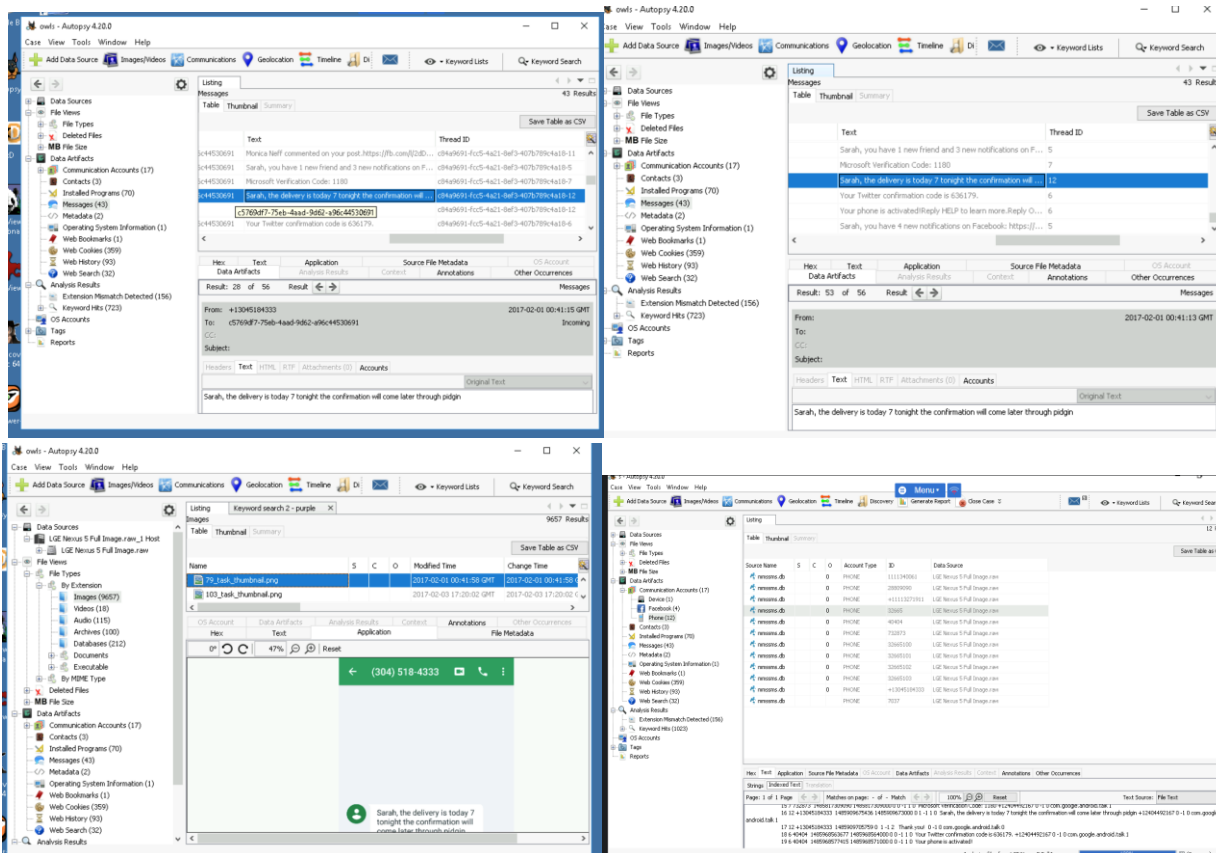
Justification for Methods and Tools

Before examination, we began with hash verification to ensure the integrity of the image. The hashes matched, meaning that the image was not tampered with since the time of acquisition.

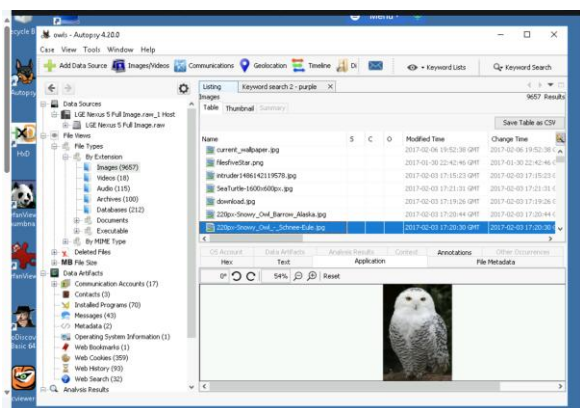
Autopsy was used for the bulk of the investigation due to its convenience and simplicity.

Autopsy ingest modules take a lot of the work out of navigating the filesystem and identifying relevant databases for messages, contacts, calls, browser history, etc. The result of ingest in Autopsy is a simple interface with all relevant artifacts easily viewable. Autopsy also allows for export to CSV, streamlining the documentation process.

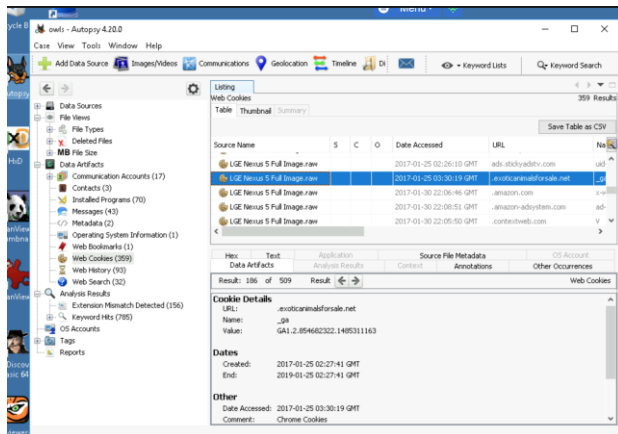
Evidence:



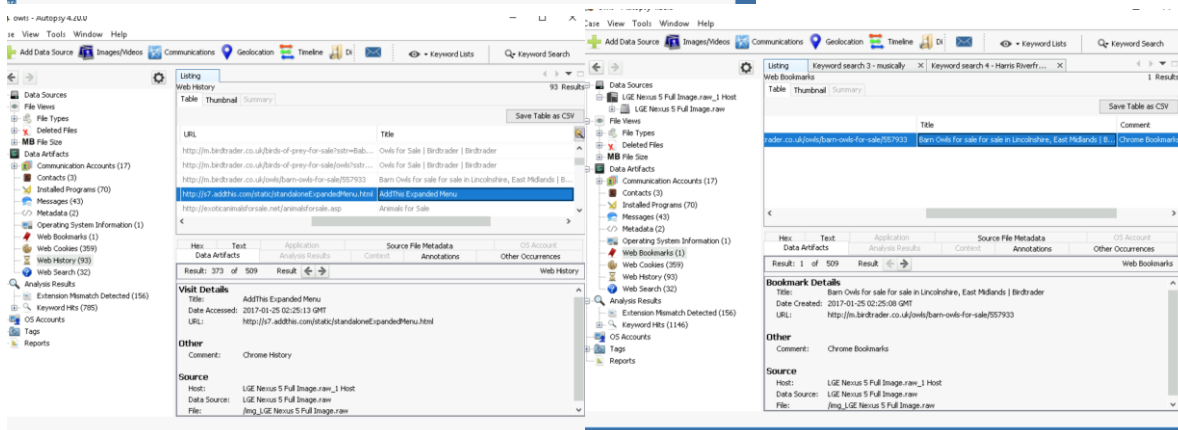
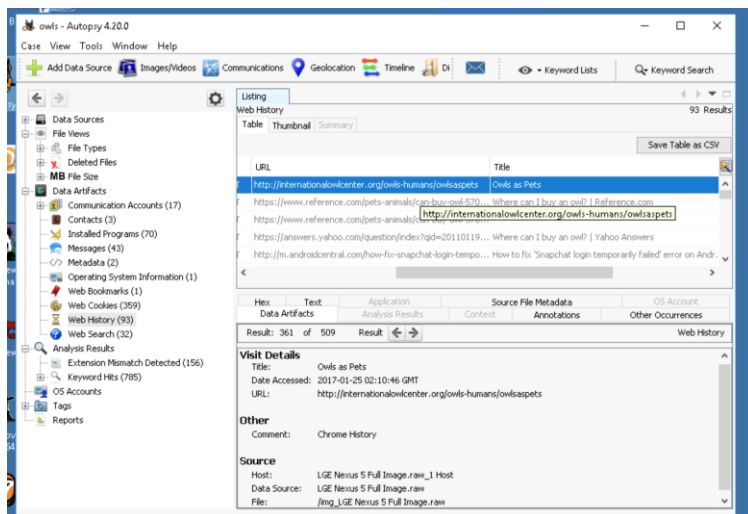
This shows Sarah getting a delivery for something on 2017-02-01 00:41:15 GMT on a messaging platform with the same number. This chat is the proof of moving to Pidgin, a universal chat client often used for encrypted or off the record messaging. Phone number of message: +13045184333 (US). There is also a twitter confirmation code meaning she might have set up a burner twitter account around the time of the talk.



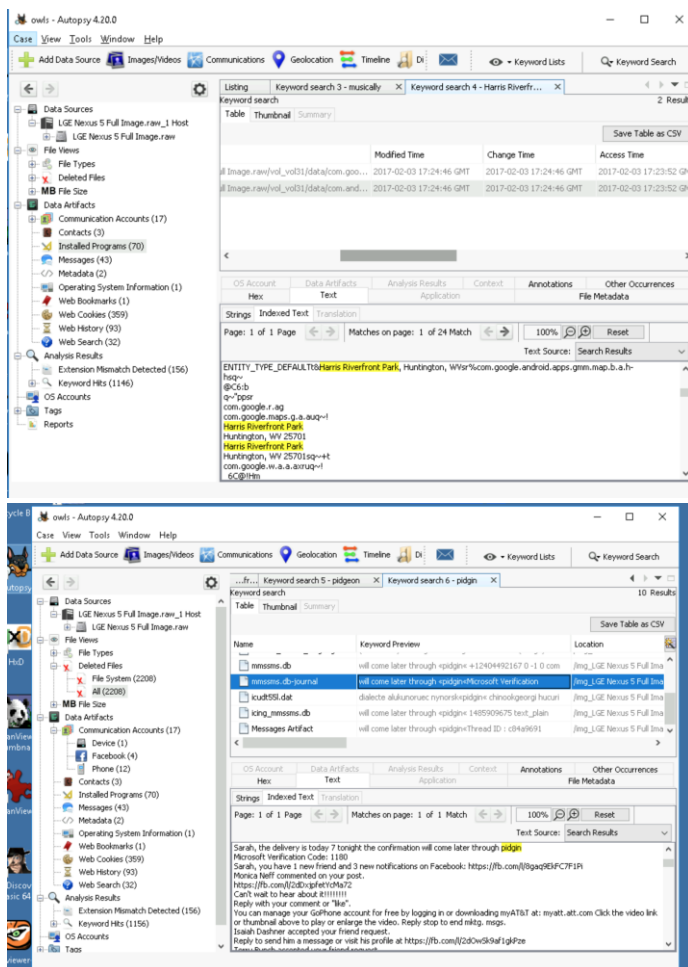
One of the animals that is possibly being bought illegally. She has multiple searches for the snowy owls and pictures of them on her phone, showing her interest in the animal.



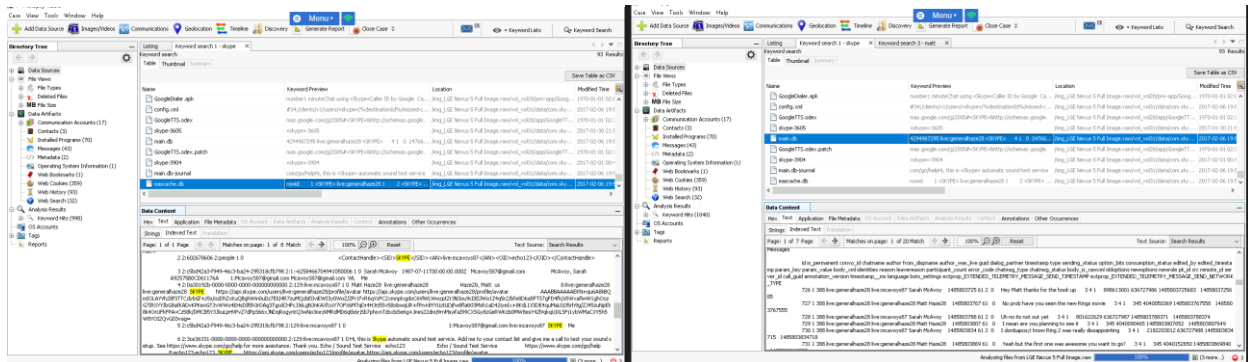
One of the sites Sarah visited was looking for exotic animals for sale called exoticanimalsforsale.net (Owls).



Searches for owning an owl as a pet as well as where to get one. She visited Barn Owls for sale in Lincolnshire, East Midlands on Bit trader (she might be making the exchange in crypto to avoid tracking the sale). The site is bookmarked too.



This location is the specific park where the hand off was arranged in her messages. The Modified Time on these location-based search records is 2017-02-03 17:24:46 GMT. This is two days after the initial "delivery at 7" message which may indicate a follow up or that the actual physical meet up happened after the first text coordination.



Incriminating evidence of a hookup with generalhaze28 (Matt Haze). The skype message from Sarah to Matt Haze explicitly states, "Hey Matt thanks for the hook up," which confirms a completed transaction between the two parties.

Summarization of evidence:

Why: She wanted an owl (proven by the Alaska Snowy Owl photos and pet care PDFs).

How: She used the SMS for initial contact (data artifact found during search), then shifted to Pidgin for "off the record" confirmation.

Who: Sarah McAvoy and her contact at +1-304-518-4333.

Chrome browser history shows research regarding purchasing an owl as a pet. The following table shows a timeline along with associated activity, exported from Autopsy.

Time (UTC)	Title / Activity	URL or Host	Significance
2017-01-25 02:10:46	“Owls as Pets”	internationalowlcenter.org/owls-humans/owlsaspets	Research into owls as pets.
2017-01-25 02:10:56	“Where can I buy an owl?”	reference.com	Direct acquisition-related research.
2017-01-25 02:11:47	“Where can I buy an owl?”	answers.yahoo.com	Direct acquisition-related research.
2017-01-25 02:20:22	“Where can you find baby owls for sale? Are owls legal to keep as pets?”	quora.com	Research into both availability and legality.
2017-01-25 02:22:44	Tawny owls for sale	m.birdtrader.co.uk	Marketplace listing activity.
2017-01-25 02:22:57	Bonding pair Asian Wood Owls for sale	m.birdtrader.co.uk	Marketplace listing activity.
2017-01-25 02:23:10	Owls for sale / baby owl search	m.birdtrader.co.uk	Marketplace search activity.
2017-01-25 02:23:27	Barn owls for sale	m.birdtrader.co.uk	Marketplace listing activity.
2017-01-25 02:25:20– 02:25:22	Owls for sale	m.birdtrader.co.uk	Repeated marketplace activity.
2017-01-25 02:26:01– 02:27:33	Exotic animals for sale; search query Owl pages 1–3	exoticanimalsforsale.net	U.S.-oriented exotic-animal marketplace searching.

Relevant Findings

Based on the forensic analysis, the relevant findings clearly show Sarah McAvoy's illegal owl acquisition. It was confirmed that the device belonged to Sarah through two linked accounts, a Google account and a Sype account, establishing her as the sole user of the device.

Starting on 25 January 2017, Sarah conducted extensive research through Chrome looking for ways to buy an owl. She searched: "Owls as Pets" and "Where can I buy an Owl?" She was also browsing multiple exotic animal markets, including birdtrader.co.uk and exoticanimalsforsale.net. This demonstrates clear intent to locate and purchase an owl. Around the same time, Sarah also browsed Amazon for bird products such as bedding, suggesting that she was prepared to buy an animal.

Communications through Skype are also relevant in proving Sarah's intent to purchase an owl. Messages between herself and Matt Haze between January 30th and 31st contained suspicious language, where Sarah says, "thanks for the hook up" and references to a meet up. On 31 January, Sarah received an SMS from the number 304-518-4333, letting her know that a delivery was scheduled for that evening and that she would receive further confirmation through Pidgin, an encrypted messaging agent. Sarah then responded within 30 seconds, indicating that she was expecting a message.

Finally, on 3 February 2017, multiple photos of snowy owls were saved to the same device, and location data pointed to a specific park, supporting the idea that a physical handoff took place. Considering all the evidence, these findings strongly support the conclusion that Sarah McAvoy knowingly researched, coordinated, and completed the illegal purchase of an owl.

Conclusion

Based on the LGE Nexus 5 Full Image, Sarah McAvoy researched, coordinated, and completed the illegal purchase of an owl. The objectives of this analysis were to identify the user of the machine, search for evidence related to the research or illegal purchase of an owl and artifacts related to other individuals in an illegal sale of an owl and create a timeline for the purchase.

To establish these findings, we began by verifying the MD5 and SHA-1 hashes of “LGE Nexus 5 Full Image.raw” against the hashes in the Magnet metadata. Then, uploaded the image to Autopsy to review the communication accounts, contacts, installed programs, messages, web bookmarks, cookies, and chrome browser history. As seen through our evidence and relevant findings, Sarah researched where she could buy an owl on two separate websites, answers.yahoo.com and quora.com on 25 January 2017 between 02:11:47 and 02:20:22. Sarah proceeded to visit m.birdtrader.co.uk on multiple occasions between 02:22:44 and 02:25:22 in pursuit of researching the marketplace listings. Finally, Sarah engaged with SMS for initial contact, then shifted to Pidgin for "off the record" confirmation, with +1-304-518-4333, where she arranged the handoff of the owl. The final communication and handoff occurred on 3 February 2017 at 17:24:46.

Combing our evidence together, Sarah McAvoy researched, coordinated, and completed the illegal purchase of an owl.